

Design Verification Plan and Report (virtual test) — VBF-T1R1FLASH-DVPR-01

Case: next-generation BEV sedan battery · Generation date: 2026-08-25

Verification items (all from bda simulation; sources file:key)

#	Item	Condition	Result	Determination	Source
1	1C discharge capacity	1C CC to 2.5 V, 25 °C, DFN	4.936 Ah	informational (nominal 5.0 Ah)	cell/final_1c_dfn.json:capacity_ah
2	Discharge energy	trapezoid $\int V \cdot I \, dt$	17.51 Wh	—	cell/final_energy.json:energy_wh
3	Gravimetric energy density	contract caliber (electrolyte excluded)	459.41 Wh/kg	PASS vs ≥ 392.61	cell/final_energy.json:energy_density_wh_kg
4	Volumetric energy density	contract caliber	893.5 Wh/L	informational	cell/final_energy.json:energy_density_wh_l
5	4C fast-charge temperature rise	4C charge, 45 °C ambient, lumped thermal, DFN	$T_{\max}$ 323.39 K (50.2 °C)	PASS vs ≤ 333.15 K	cell/archA_h150_f4b_4c.json:T_max_K
6	4C fast-charge plating	same + plating module; anode potential ≥ 0 V	anode min +0.0114 V, plated=false	PASS	cell/archA_h150_f4b_4c.json:anode_potential_v
7	Cooling robustness	4C at $h=120 \text{ W/m}^2 \cdot \text{K}$	$T_{\max}$ 324.55 K, anode min +0.0123 V	PASS (margin preserved)	cell/archA_h120_f4b_4c.json
8	Voltage window	parameter set	2.5–4.2 V	PASS	OKane2022 params
9	Overcharge to 4.7 V	0.5C charge to cutoff+0.5 V (4.7 V), then TR ODE	overcharge $T_{\max}$ 299.12 K; triggered=false	PASS (no thermal runaway)	cell/final_oc.json:T_max_K; cell/final_tr.json:triggered
10	DC resistance	$(OCV - V_{@10\% \text{ DOD}})/I$	20.3 mΩ	informational	cell/final_energy.json:dcr_ohm

Items NOT covered (beyond pure-simulation boundary, require physical experiment)

Nail penetration, crush, drop, cycle life / calendar aging (aging model not part of this task's criteria), rate-pulse internal resistance, electrolyte oxidation voltage-window endorsement (real_compute=false), production-level formation tuning.

Conclusion

All simulated verification items PASS vs the entry-0 contract ($ED \geq 392.61 \text{ Wh/kg}$; $T_{\max} \leq 333.15 \text{ K}$; plated=false; overcharge→TR triggered=false). Uncovered items listed above are the honest boundary of this virtual-design package.